

# **ELEC50001 – Circuits and Systems**

## **SAMPLE PAPER 1**

Answer ALL questions.

There are THREE questions on the paper.

Question ONE counts for 50% of the marks, other questions 25% each

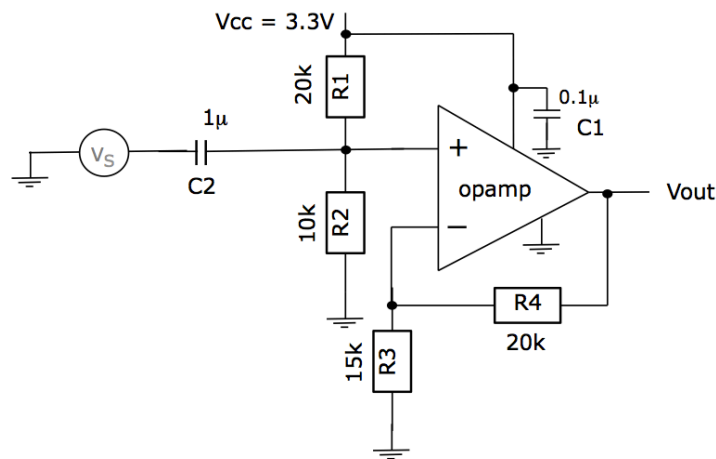
Time allowed: 2 hours

**Information for Candidates:**

The following notation is used in this paper:

1. Unless explicitly indicated otherwise, digital circuits are drawn with their inputs on the left and their outputs on the right.
2. Within a circuit, signals with the same name are connected together even if no connection is shown explicitly.
3. The notation  $X[2:0]$  denotes the three-bit number  $X_2$ ,  $X_1$  and  $X_0$ . The least significant bit of a binary number is always designated bit 0.

1. (a) *Figure 1.1* shows an amplifier circuit using an op-amp operating at 3.3V and has a gain-bandwidth product of 1MHz. It amplifies the input signal  $V_S$  and produces an output voltage  $V_{out}$ .
- (i) Derive from first principle the gain of this amplifier for both the DC and AC components of  $V_S$ . You must show your derivation clearly. [4]
- (ii) Given that  $V_S$  is a sine wave signal at 20kHz with a peak-to-peak voltage of 1V, sketch and explain the expected waveform at  $V_{out}$ . [3]
- (iii) State with justifications what happens to the gain of the amplifier if the frequency of  $V_S$  is changed to 1Hz and to 1MHz. [3]



*Figure 1.1*

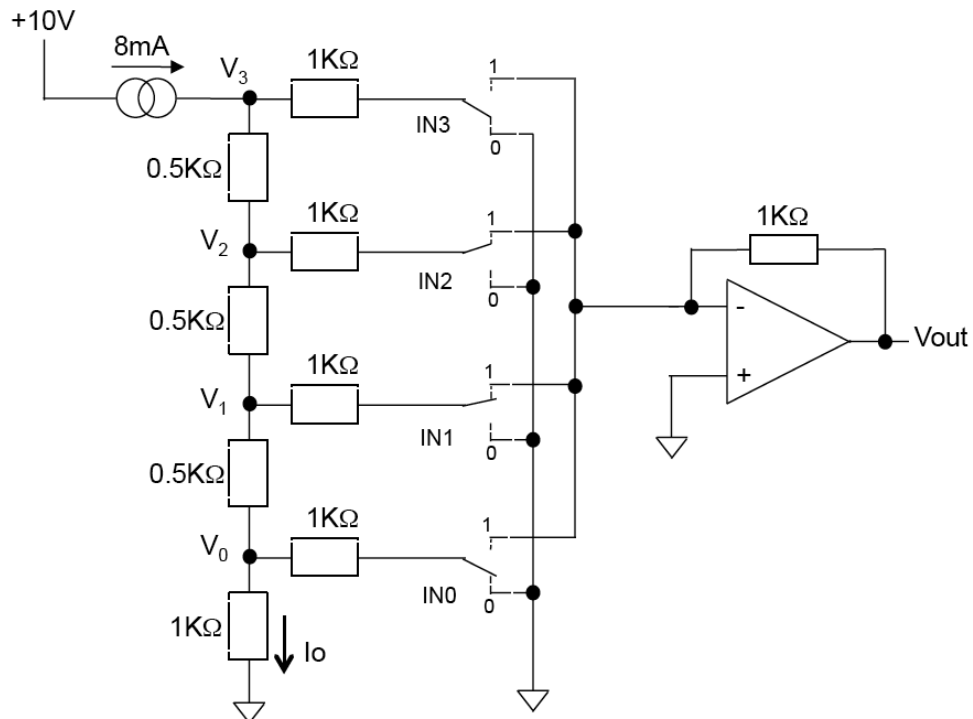
- (b) (i) Explain the meaning of the following specifications for a digital-to-analogue converter (DAC): resolution, linearity error, monotonicity and settling time. Give one example where the monotonic behaviour of a DAC is important.

[4]

- (ii) *Figure 1.2* shows a circuit for a R-2R 4-bit DAC with input  $IN[3:0] = 4'b0110$ . The resistor network is supplied via a 8mA ideal current source. The digital inputs  $IN[3:0]$  control the four 2-way electronic switches. The output  $V_{out}$  is driven by an ideal operational amplifier.

Derive the values for the current  $I_0$ , and voltages  $V_0$ ,  $V_2$  and  $V_{out}$ .

[4]



*Figure 1.2*

(c) *Figure 1.3* shows a 4k x 4 RAM chip with address bus ADR, data bus DATA, a write enable signal WR, a chip select signal CS and a synchronising clock CLK.

(i) Specify the size of the address and data buses for the RAM chip.

[2]

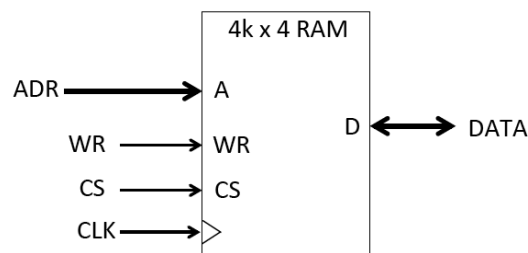
(ii) An 8k x 8 bit RAM module is required in a microprocessor system with a 16-bit address bus and an 8-bit data bus. The starting address of the RAM module is 16'h8000.

a) Design, in the form of a schematic diagram, a circuit using four 4k x 4 RAM chips shown in *Figure 1.3* to implement the 8k x 8 RAM module.

[5]

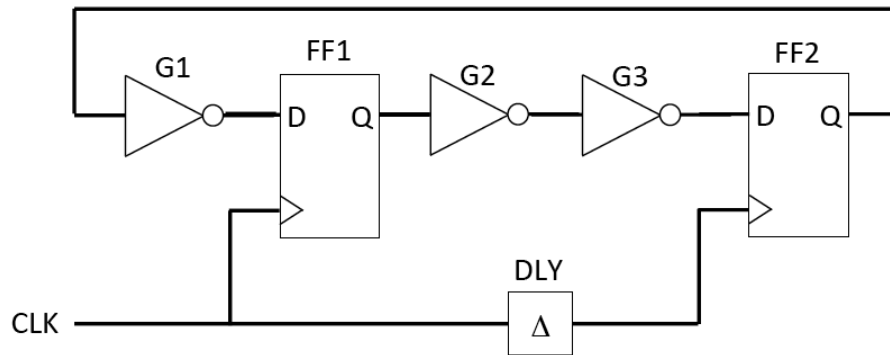
b) Derive the Boolean expression for the chip select signals required for each of the 4k x 4 RAM chip.

[3]



*Figure 1.3*

- (d) Consider the circuit shown in *Figure 1.4*. The propagation delay  $t_d$  of the inverters G1 to G3, and the clock-to-Q delay  $t_{cq}$  of the flip-flops FF1 and FF2, are in the range of 0.5ns to 1.0ns. The setup and hold times of the flip-flops FF1 and FF2 are 1.3ns and 1.1ns respectively. The clock to FF2 is driven through a delay element DLY with a delay of  $t_{dly}$ .
- (i) Assuming that  $t_{dly} = 0$ ns and using only the setup time constraint, calculate the maximum operating frequency of the clock signal CLK. [4]
- (ii) Assuming that  $t_{dly} = 0$ ns, show that there is a hold time violation for FF1. [3]
- (iii) Delaying the clock to FF2 can eliminate the hold time violation. Show that the hold time violation is eliminated if the delay  $t_{dly} > 0.1$  ns. [3]



*Figure 1.4*

(e) Figure 1.5a shows the Verilog specification of a finite state machine FSM with four states that uses one-hot encoding.

- (i) What is one-hot encoding? Briefly explain why one-hot encoding is particularly suitable for FPGA implementation. [4]
- (ii) Construct the state transition diagram for this FSM. [4]
- (iii) Hence or otherwise, complete the timing diagram shown in Figure 1.5b. [4]

```

module FSM (x, a, clk, rst);
input      a, clk, rst;
output     x;

// define states - one-hot encoding
parameter  NSTATE = 4;
parameter  S_0 = 4'b0001;
parameter  S_1 = 4'b0010;
parameter  S_2 = 4'b0100;
parameter  S_3 = 4'b1000;

reg  [NSTATE-1:0]  state;
reg  x;
wire  a, clk, rst;

// specify state machine
always @ (posedge clk)
    if (rst == 1'b1) begin
        state <= S_0;
        x <= 1'b0; end
    else
        case (state)
        S_0: if (a==1'b1) begin
                state <= S_0; x <= 1'b0; end
            else begin
                state <= S_1; x <= 1'b0; end
        S_1: if (a==1'b1) begin
                state <= S_2; x <= 1'b0; end
            else begin
                state <= S_3; x <= 1'b1; end
        S_2: begin state <= S_3; x <= 1'b1; end
        S_3: if (a==1'b1) begin
                state <= S_1; x <= 1'b1; end
            else begin
                state <= S_0; x <= 1'b0; end
        default: begin state <= S_0; x <= 1'b0; end
        endcase
    endmodule

```

Figure 1.5a

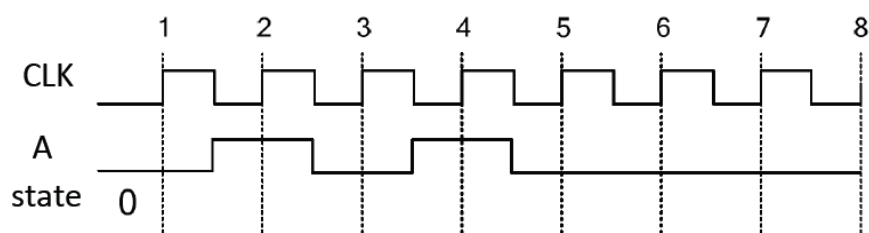


Figure 1.5b

2. Figure 2.1 (a) shows a digital circuit that computes the approximate square root of an 8-bit number  $P[7:0]$  and produces a 4-bit answer  $X[3:0]$ . The circuit consists of a squaring circuit, a digital comparator and a finite state machine that implements the successive approximation algorithm. The digital comparator output *HIGHER* is high if the value at its input *A* is larger than its input *B*. The FSM is controlled by the clock signal *CLK*. It starts the computation when *LOAD* goes from low to high and the *DONE* signal then goes low until the computation is completed and  $X[3:0]$  contains the valid result.

- (a) Assuming that  $P[7:0] = 8'h54$ , complete the timing diagram shown in figure 2.1 (b).

[7]

- (b) Hence or otherwise, explain how this circuit is able to compute the square root of the input. State any assumptions used.

[8]

- (c) Design in Verilog HDL the finite-state machine SAR FSM.

[10]

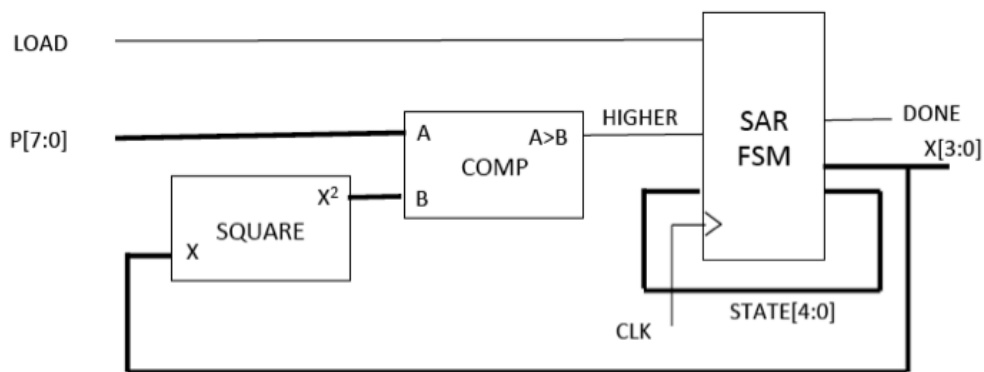


Figure 2.1 (a)

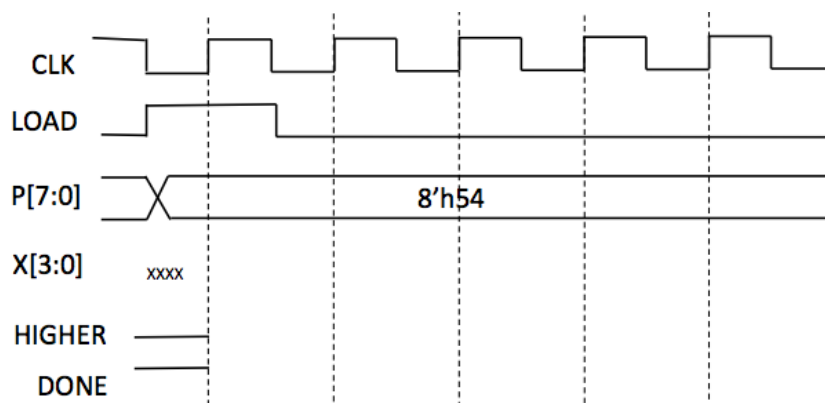


Figure 2.1 (b)

3. Figure 3.1 shows a circuit consisting of a 2-bit up counter CTR2, a 3-bit counter CTR3 which counts up when EN is high, a 2-input AND gate, a 3-bit register FF which is enabled when G2 is high and a 3-bit binary subtractor SUB. The circuit is clocked by the signal CLK, and has an input A and an output Z[2:0].

(a) Specify this circuit in Verilog HDL.

[10]

(b) Assuming that both counters and the register are initially reset, complete the timing diagram in figure 3.2 by showing the values of K[1:0], Q[2:0], Y[2:0] and Z[2:0].

[15]

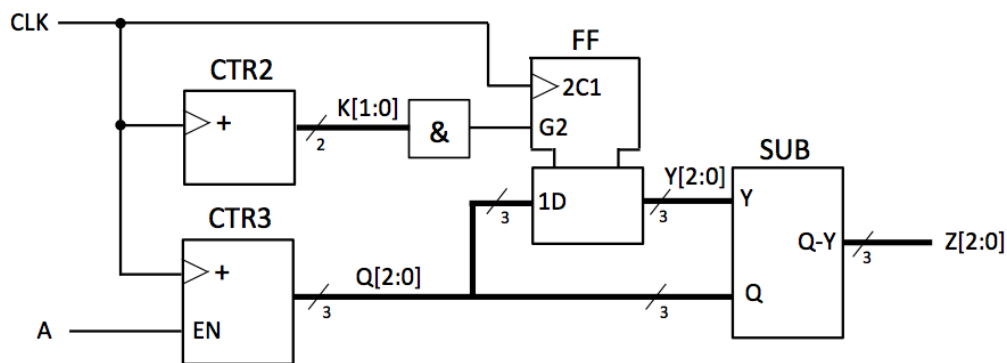


Figure 3.1

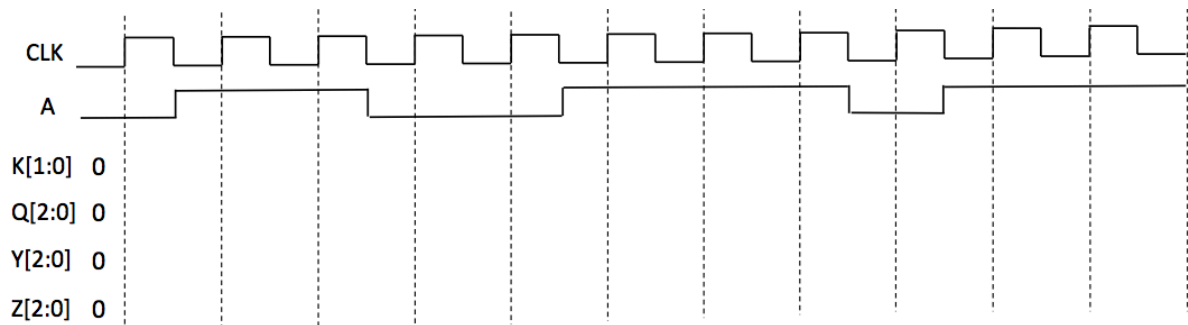


Figure 3.2